

On the Construction of Configurations n_3

Inaugural Dissertation

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by **Ernst Steinitz of Breslau**

Opponents: Herr Hans Räck, cand. phil.; Herr Reinhold Pyrkosch, stud. phil.

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Dedicated to his highly honored teacher

Professor Dr. Rosanes

by the author.

The construction of configurations n_3 has so far been treated in detail only in special cases. Martinetti recognized that, in general, it leads to a problem of the second degree, for example to the problem of determining the double elements of two incident point rows, and he also gave an indication of the path leading to this.¹ The subject is, however, interesting enough to deserve a more detailed treatment. Such a treatment will be given below.

Methods for the linear production of the configurations n_3 will then be attached to this. Schroeter² was the first to give such a procedure for a special class of configurations; it can also be applied to many other configurations. We shall see that for most configurations n_3 a linear construction is possible. The construction methods that we shall give cannot always be carried out, but they fail only in a few cases.

§1. Definition

A plane configuration n_3 (Cf. n_3) is a plane geometric figure consisting of n lines and n points so situated with respect to one another that every line contains three points and three lines pass through every point.

We denote the points of the configuration by n signs, for example by the numerals $1, 2, \dots, n$, and write these numerals in n columns, each of which is to represent a line of the configuration, by containing the three numerals that denote the points lying on the line. In this way we arrive at a schematic representation of the configuration. Each configuration can be denoted in many ways. Apart from the fact that it is immaterial how we arrange the columns of the scheme and the elements (numerals) inside any one column, the order in which we denote the points of the configuration by the numerals $1, 2, \dots, n$ is arbitrary. We shall therefore regard two configurations as the same if the scheme of one passes into that of the other by a substitution of the signs $1, 2, \dots, n$.

From the definition of configurations n_3 the following properties result for their notation:

1. Every column contains three elements.
2. Every element occurs in three columns.

¹Martinetti, "Sulle configurazione piane 4_3 ," Section 3, Annali di matematica pura ed applicata, Series II, vol. 15.

²Schroeter, "Über lineare Constructionen zur Herstellung der Configurationen n_3 " and "Über die Bildungsweise und geometrische Construction der Configurationen 10_3 ," Nachrichten der Königl. Gesellschaft der Wissenschaften zu Göttingen, 1888 and 1889.

3. A column may not contain one and the same element more than once.
4. Two columns may not agree in more than one element.

Every grouping of n elements into n columns that has these four properties will be called the scheme of a configuration n_3 or, when no misunderstanding is possible, simply a configuration n_3 ; the elements will also be called configuration-points and the columns configuration-lines, even if the scheme does not owe its origin to a geometric configuration. Conversely, starting from an arbitrary such scheme, we shall now investigate how geometric configurations corresponding to the given scheme can be produced.

§2. General decomposition into horizontal rows

A property that belongs not only to configurations n_3 but to all configurations n_μ helps us to a construction valid for all configurations n_3 ; we shall therefore prove the presence of this property in general for configurations n_μ .³

Theorem. In the scheme of every configuration n_μ , the elements inside the columns can be arranged in such a way that every horizontal row⁴ contains each of the elements $1, 2, \dots, n$ once, and hence only once.

To prove this theorem, we first show that one can associate the points and lines of the configuration in such a way that to each point there corresponds a line passing through it. Suppose that such an association has already been made with a number of points and lines. Let the complex of these lines be A , the complex of these points $\mathfrak{A}$, the complex of the remaining lines B , and the complex of the remaining points $\mathfrak{B}$. Then to every point of $\mathfrak{A}$ there corresponds a line of A passing through it, and our task plainly consists in enlarging the complex A so that it comprises all lines of the configuration. Then $\mathfrak{A}$ will contain all points of the configuration, and the complexes B and $\mathfrak{B}$ will disappear.

If a line b of B contains a point b of $\mathfrak{B}$, then we can associate b and b with one another. The point b thereby passes from $\mathfrak{B}$ into $\mathfrak{A}$, and the line b passes from B into A . This procedure is not applicable if all lines of B contain only points of $\mathfrak{A}$.

In that case we denote by a_0 all points of $\mathfrak{A}$ that lie on a line of B ; when distinction is needed we attach a second index. The corresponding lines, which must all belong to A , are denoted by α_0 . The points of $\mathfrak{A}$ lying on these lines but not occurring among the points a_0 are denoted by a_1 , their corresponding lines by α_1 ; the points of $\mathfrak{A}$ lying on these lines and not belonging to a_0 or a_1 are denoted by a_2 , their corresponding lines by α_2 , and so on. Since n is finite, we must finally arrive at points a_k whose corresponding lines α_k contain only points of $\mathfrak{A}$ already included among $a_0, a_1, \dots, a_k$.

We divide the complex $\mathfrak{A}$ into two complexes $\mathfrak{A}^0$ and $\mathfrak{A}'$, the first containing all points $a_0, \dots, a_k$, while the other consists of the remaining points of $\mathfrak{A}$. If the lines corresponding to the points of $\mathfrak{A}^0$ are assigned to a complex A^0 , and the lines corresponding to the points of $\mathfrak{A}'$ to a complex A' , then A^0 and A' together form the complex A .

Assume that a line of the complex A^0 , say α_i with $i \leq k$, contains a point of the complex $\mathfrak{B}$, say b' . If a'_i is the point corresponding to α_i , then it is contained in at least one of the lines α_{i-1} ; let α'_{i-1} be such a line, and let a'_{i-1} be the point corresponding to it. Then a'_{i-1} is

³By a configuration n_μ we here mean a grouping of n elements in n columns in which every column (line) contains μ elements (points), every element occurs in μ columns, and which also satisfies the third and fourth of the conditions set up for configurations n_3 .

⁴We imagine the columns of the configuration written side by side so that beside the k -th element of any column stands the k -th element of the next column.

contained in at least one of the lines α_{i-2} ; let α'_{i-2} be such a line. Continuing in this way we arrive at a sequence of points and lines

$$b', \alpha_i, a'_i, \alpha'_{i-1}, a'_{i-1}, \dots, \alpha'_0, a'_0, b'',$$

whose last member b'' is one of the lines of the complex B containing the point a'_0 . This sequence has the following properties. Every point, with the exception of b' , is associated with the preceding line and lies on it. But every point also lies on the following line. We can therefore change the association by making each point correspond to the following line. The points that previously belonged to $\mathfrak{A}$ remain in it; likewise all lines previously in A remain in A . But now the point b' , to which the line α_i is associated, passes from $\mathfrak{B}$ into $\mathfrak{A}$, and the line b'' , to which the point a'_0 belongs, passes from B into A .

It remains only to consider the case where no line of the complex A^0 contains a point of the complex $\mathfrak{B}$. The points of $\mathfrak{B}$, since they also lie on no line of B , can then occur only on the lines of the complex A' . But this too is impossible. The points of $\mathfrak{A}'$, as follows immediately from their definition, occur neither on the lines of B nor on those of A^0 ; they are therefore confined to the lines of A' . Since $\mathfrak{A}'$ contains just as many points as A' contains lines, while each point occurs μ times in the configuration and each line contains μ points, it follows that the lines of A' contain only the points of $\mathfrak{A}'$. Thus the complex B , and likewise $\mathfrak{B}$, is not present at all; the desired association of lines and points has already been reached.

We now place, in each of the columns denoting the lines of the configuration, that element in first position which denotes the point associated with the line. The first horizontal row of the configuration scheme will then contain every element once. The scheme that remains after removal of this row plainly denotes a configuration $n_{\mu-1}$, for which the theorem proved in general for configurations n_μ must again hold. We can therefore arrange the elements inside the columns of this scheme so that its first horizontal row, the second of the original scheme, contains every element once. It is immediate that the elements of the remaining horizontal rows, which always form a configuration, can be arranged so that each of those rows also contains every element once. The theorem stated above is thereby completely proved.

We apply the method just given to the following configuration 20₃:

Cf. 20_3:

1	1	1	2	2	2	3	3	4	4	5	5	6	6
3	4	8	9	11	15	5	15	7	12	7	10	7	8
10	9	17	16	14	18	13	19	13	16	14	18	20	10
6	8	9	11	11	12								
14	12	17	15	16	13								
19	19	20	20	17	18								

If we associate with the listed lines the points

$$1, 9, 2, 3, 19, 4, 16, 5, 10, 7, 8, 14, 17, 11, 12,$$

then these lines form the complex A and the corresponding points form $\mathfrak{A}$. The complex B consists of the lines

$$\begin{array}{cccccc} 1 & 1 & 2 & 3 & 11 \\ 3 & 4 & 11 & 12 & 16 \\ 10 & 9 & 14 & 19 & 17, \end{array}$$

and $\mathfrak{B}$ consists of the points 6, 13, 15, 18, 20.

All points lying on the lines of B belong to $\mathfrak{A}$; these are the points a_0 . Among the points a_1 one now has the points 5 and 7. The points 13, 18, 15, 6 belonging to $\mathfrak{B}$ lie respectively on the

lines

$$\begin{array}{cccc} 4 & 2 & 11 & 6 \\ 7 & 5 & 15 & 8 \\ 13 & 18 & 20 & 10. \end{array}$$

If we associate them with those lines, and instead associate with the points 4, 2, 11, 8 the lines

$$\begin{array}{cccc} 1 & 2 & 11 & 8 \\ 4 & 11 & 16 & 12 \\ 9 & 14 & 17 & 19, \end{array}$$

then B contains only the line $\begin{array}{c} 1 \\ 3 \\ 10 \end{array}$, and $\mathfrak{B}$ only the point 20. Further, one has

$$\begin{array}{lcl} a_0 & : & 1, 3, 10, \\ a_1 & : & 5, 8, 13, 17, 18, \\ a_2 & : & 2, 4, 7, 9, 12, 14, 15, 19, \\ a_3 & : & 6, 11, 16. \end{array}$$

The point 20 lies on the line corresponding to point 17; point 17 lies on the line corresponding to point 1; and point 1 lies on the line belonging to B . If we now associate with the points 20, 17, 1 the lines

$$\begin{array}{ccc} 9 & 1 & 1 \\ 17 & 8 & 3 \\ 20 & 17 & 10, \end{array}$$

then the desired association has been completely carried out.

Putting in each column, in first place, the element denoting the point associated with the corresponding line, and then deleting the first horizontal row of the scheme, one obtains three polygons

$$1, 9, 17, 16, 2, 15, 3, 10, 8; \quad 4, 7, 14, 11, 20, 6, 19, 12; \quad 5, 13, 18,$$

which compose the configuration.

Further decompositions of the configuration can easily be derived from this one. If we also arrange the second and third horizontal rows so that each contains all elements – and, if arrangements are regarded as the same when they can be carried into one another by interchanging columns and horizontal rows, this is possible in four ways – then, after deleting the second or third horizontal row, new representations of the configuration as a cycle of polygons result. The four essentially different possible arrangements are the following:

I.

$$\begin{array}{cccccccccccccccccccc} 4 & 20 & 11 & 9 & 18 & 19 & 1 & 6 & 17 & 13 & 5 & 2 & 15 & 7 & 14 & 8 & 16 & 3 & 12 & 10 \\ 1 & 9 & 17 & 16 & 2 & 15 & 3 & 10 & 8 & 4 & 7 & 14 & 11 & 20 & 6 & 19 & 12 & 5 & 13 & 18 \\ 9 & 17 & 16 & 2 & 15 & 3 & 10 & 8 & 1 & 7 & 14 & 11 & 20 & 6 & 19 & 12 & 4 & 13 & 18 & 5 \end{array}$$

II.

$$\begin{array}{cccccccccccccccccccc} 4 & 20 & 11 & 9 & 18 & 19 & 1 & 6 & 17 & 16 & 8 & 14 & 7 & 15 & 2 & 5 & 13 & 3 & 12 & 10 \\ 1 & 9 & 17 & 16 & 2 & 15 & 3 & 10 & 8 & 4 & 12 & 19 & 6 & 20 & 11 & 14 & 7 & 5 & 13 & 18 \\ 9 & 17 & 16 & 2 & 15 & 3 & 10 & 8 & 1 & 12 & 19 & 6 & 20 & 11 & 14 & 7 & 4 & 13 & 18 & 5 \end{array}$$

III.

$$\begin{array}{cccccccccccccccccccc} 4 & 20 & 11 & 9 & 18 & 19 & 1 & 6 & 17 & 16 & 8 & 14 & 7 & 15 & 2 & 5 & 13 & 10 & 12 & 3 \\ 1 & 9 & 17 & 16 & 2 & 15 & 3 & 10 & 8 & 4 & 12 & 19 & 6 & 20 & 11 & 14 & 7 & 5 & 18 & 13 \\ 9 & 17 & 16 & 2 & 15 & 3 & 10 & 8 & 1 & 12 & 19 & 6 & 20 & 11 & 14 & 7 & 4 & 18 & 13 & 5 \end{array}$$

IV.

4	20	11	9	18	19	1	6	17	13	5	2	15	7	14	8	16	10	12	3
1	9	17	16	2	15	3	10	8	4	7	14	11	20	6	19	12	5	18	13
9	17	16	2	15	3	10	8	1	7	14	11	20	6	19	12	4	18	13	5

Deleting in these schemes first the second and then the third horizontal row gives the following eight new decompositions of the configuration into polygons:

- 1) 1 10 5 14 19 3 13 7 6 8 12 18 15 20 17; 2 11 16 4 9.
- 2) 1 3 5 7 20 9 16 12 13 4; 2 14 6 10 18; 8 19 5 11 17.
- 3) 1 10 5 7 20 17; 2 14 6 8 19 3 13 4 9; 11 16 12 18 15.
- 4) 1 3 5 14 19 15 20 9 16 4; 2 11 17 8 12 13 7 6 10 18.
- 5) 1 10 18 15 11 16 12 13 4 9 2 14 6 8 19 3 5 7 20 17.
- 6) 1 3 13 7 6 10 5 14 19 15 20 9 16 4; 2 11 17 8 12 18.
- 7) 1 10 18 15 20 17; 2 11 16 4 9; 3 5 14 19; 6 12 13 7.
- 8) 1 3 13 4; 2 14 6 10 5 7 20 9 16 12 18; 8 19 15 11 17.

Further decompositions can be derived from these in the same way.

§3. Closed traversals

The proof just given is general and also includes the case where the configuration n_μ decomposes into several configurations. The latter case occurs, for example, whenever the complex A' disappears.

If ν horizontal rows are omitted from the configuration n_μ , then a configuration $n_{\mu-\nu}$ remains, which of course may also decompose. For $\nu = \mu - 2$ one obtains a polygon or a cycle of polygons, each side of which still contains $\mu - 2$ additional points of the configuration, while through each vertex there still pass $\mu - 2$ lines. We see from this that every configuration n_μ can be traversed in several closed circuits in such a way that every line occurs once as a side and every point once as a vertex.

If the given configuration n_μ decomposes, then every configuration n_ν contained in it, with $\nu < \mu$, must also decompose. But the converse of this theorem is not valid. A configuration n_μ may be completely connected, so that one can pass from any one of its points to any other by proceeding along configuration-lines, and nevertheless every configuration $n_{\mu-1}$ contained in it may decompose. Such cases already occur for configurations n_3 . Thus not every connected configuration n_3 can be traversed in a single closed circuit; rather, for any given number k one can form configurations n_3 that do not decompose but that cannot be traversed in fewer than k circuits.

§4. Arrangement of the complexes

In the following investigations, where we deal only with configurations n_3 , we assume that they do not decompose.

If the elements in the scheme of a configuration n_3 are arranged so that every horizontal row contains every element, then this property is preserved when the horizontal rows are interchanged; after such an interchange the scheme also denotes the same configuration as before. We now put an arbitrarily chosen horizontal row in first place; the two others denote a number of polygons.

To make this still clearer, and keeping the order of the elements within the columns unchanged, we arrange the columns as follows. We choose any column as the first, and determine the following columns in such a way that each second element of a column is identical

with the third of the preceding one. This procedure can be continued until we arrive at a column whose third element is identical with the second of the initially chosen column. The totality of columns, or lines, obtained in this way is called a complex of the configuration. If the complex does not yet comprise all columns of the configuration, we proceed to form a second complex by again taking an arbitrary one of the remaining columns and determining each following column in the same way as in the first complex. If the configuration is still not exhausted by this second complex, we form a third, fourth, and so on. That such a mode of formation is possible follows at once from the fact that every horizontal row of the scheme contains every element once.

If the first horizontal row of the configuration is omitted, one polygon remains from each complex. The second, and likewise the third, horizontal row of the complex at the same time gives the order in which the vertices of the polygon follow one another. This property is preserved when the columns of a complex are cyclically interchanged; no other interchange is permitted.

Let the number of complexes be k . We arrange them as follows. First we determine an arbitrary complex as the last one; call it C_k , its first horizontal row R_k , and the polygon represented by the two other rows P_k . If the configuration does not consist only of this complex, then not all elements of P_k can be contained in R_k ; at least one of them must occur in the first row of another complex. Otherwise the complex C_k would by itself already form a configuration.

We place such a complex in the next-to-last position and make a cyclic interchange of its columns so that the first element of the last column belongs to the elements contained in P_k . Call the complex thus determined C_{k-1} ; let R_{k-1} be its first horizontal row and P_{k-1} the polygon contained in it. If $k > 2$, not all points of P_{k-1} and P_k can be represented in the rows R_{k-1} and R_k ; at least one of them must occur in the first row of another complex. We choose such a complex as C_{k-2} and, by cyclically interchanging its columns, arrange that the element of P_{k-1} or P_k contained in R_{k-2} becomes the first element of the last column of C_{k-2} . It is clear how this procedure is to be continued, and that it leads to complexes $C_1, C_2, \dots, C_k$ with first horizontal rows $R_1, R_2, \dots, R_k$ and polygons $P_1, P_2, \dots, P_k$, having the property that the last point of R_i ($i < k$) occurs as a point of the second and third row only in one of the following polygons $P_{i+1}, \dots, P_k$.

§5. Normal notation

After the columns of the configuration have been arranged in this way, we replace the elements of the first horizontal row of the scheme, in order, by the signs $t_1, t_2, \dots, t_n$, and those of the second row by $p_1, p_2, \dots, p_n$. Each element of the third row is to be replaced by the sign that was introduced for the same element in the second row. We further denote by r_i ($i = 1, 2, \dots, n$) the column whose first element is t_i , and by p_i the column whose first element denotes the point also represented by the sign p_i . The notation introduced for elements and columns will also be used for the points and lines themselves. Thus each point and each line of the configuration receives two names.

If α_i ($i = 1, 2, \dots, k$) is the number of lines belonging to the complex C_i , we have the following scheme for the configuration:

$$\begin{array}{cccccc} \text{Cf. } n_3 & & & & & \\ t_1 & t_2 & \cdots & t_{\alpha_1} & t_{\alpha_1+1} & \cdots \\ p_1 & p_2 & \cdots & p_{\alpha_1} & p_{\alpha_1+1} & \cdots \\ p_2 & p_3 & \cdots & p_1 & p_{\alpha_1+2} & \cdots \end{array}$$

and analogously for the remaining complexes, up to the last one.

We now fix the following normal order for the points and lines of the configuration:

$$p_1, r_1, p_2, r_2, \dots, p_n, r_n.$$

This order is always being referred to when, of two elements (points or lines) of the configuration, one is said to precede or follow the other. Expressions such as “the first point,” “the last line of a polygon,” and the like are to be understood accordingly.

§6. Construction after $3n - 1$ conditions

The configuration n_3 can be regarded as a grouping of n lines $r_1, r_2, \dots, r_n$ and n points $p_1, p_2, \dots, p_n$ that arises by subjecting these initially arbitrary elements, one after another, to $3n$ conditions, not always independent, as if n lines and n points in the plane were moved until they assumed the position prescribed by the configuration. We first seek to produce a figure in which only $3n - 1$ of the $3n$ conditions are satisfied: for the time being we omit the requirement that the line r_n contain the point t_n . Notice that the omitted condition is not distinguished from the others. Since we placed an arbitrary complex last, and since any cyclic interchange was allowed among the lines of that complex, any line of the configuration can occupy the last place; hence r_n is an arbitrary line of the configuration. Since the horizontal rows of the scheme could also be interchanged arbitrarily, t_n is an arbitrary point of the line r_n .

We construct the figure in question by determining the points and lines in the normal order

$$p_1, r_1, p_2, r_2, \dots, p_n, r_n$$

and ensuring that the remaining $3n - 1$ conditions of the configuration are fulfilled. It is easy to see that this is possible. Suppose that the elements

$$p_1, r_1, p_2, r_2, \dots, p_i, r_i$$

have already been constructed. The point p_{i+1} is now to be determined.

There are two cases. If p_{i+1} is not the first point of one of the polygons $P_1, \dots, P_k$, then the configuration requires it to lie on the lines r_i, r_{i+1}, p_{i+1} . Of these lines r_{i+1} is not yet constructed, whereas r_i is; for p_{i+1} both cases are possible. Accordingly, depending on whether p_{i+1} belongs to the lines already constructed or not, the point p_{i+1} is to be placed at the intersection of the lines p_{i+1} and r_i , or it is to be taken as an arbitrary point of r_i .

If p_{i+1} is the first point of a polygon, and p_j the last point of the same polygon, then p_{i+1} must lie on the lines r_{i+1}, p_j, p_{i+1} , of which at most p_{i+1} has already been constructed. According to whether this is the case or not, p_{i+1} is to be taken arbitrarily on the line p_{i+1} , or altogether arbitrarily.

In constructing the following line r_{i+1} , two cases are again to be distinguished. If r_{i+1} is not the last line of a polygon, it must contain the points $p_{i+1}, p_{i+2}, r_{i+1}$. Of these, the first has already been found and the second has not. Depending on whether the point r_{i+1} belongs to the already constructed points or not, r_{i+1} must be the joining line of p_{i+1} and r_{i+1} , or an arbitrary line through the former point.

If r_{i+1} is the last line of one of the polygons $P_1, P_2, \dots, P_k$, and r_m is the first line of the same polygon, then r_{i+1} must contain the points p_n, r_{i+1}, p_{i+1} . The points p_n and p_{i+1} have already been determined, but r_{i+1} has not; for, in our arrangement of the configuration-lines, those points that lie on the last line of a polygon but do not occur there as vertices of that polygon must be vertices of one of the following polygons. Thus r_{i+1} must be constructed as the joining line of p_n and p_{i+1} .

Finally, if p_m is the first point of the last polygon and we construct r_n as the joining line $p_m p_n$, the constructed figure will satisfy $3n - 1$ conditions of the configuration.

§7. The last condition

In order to satisfy the last condition of the configuration as well, we seek, in the sequence of elements

$$p_1, r_1, p_2, r_2, \dots, p_n, r_n,$$

the last one that, after the preceding elements have been constructed and the last configuration-condition has been ignored, is not yet completely determined. We then try to dispose of this element in such a way that this condition also is fulfilled when the construction of the configuration is completed by the procedure described in §6. It is clear that the element sought belongs to the last polygon, as vertex or side, since the first point of this polygon, p_m , is not completely determined by the preceding elements. The element sought may be partially determined by what precedes it: if it is a point, it may be forced to lie on an already constructed line; if it is a line, it may be forced to pass through an already constructed point. It is also possible, however, that after the construction of the preceding elements it is still entirely undetermined.

This last case, which by the discussion of §6 can occur only when p_m , the first point of the last polygon, is the element sought, can be avoided. Since the point t_{m-1} must be a vertex of the last polygon, a cyclic interchange of the vertices and sides of the latter can be made so that $p_m = t_{m-1}$. Then p_m lies on the already constructed line t_{m-1} .

Let the element sought be a point p_i , with $i > m$. If, in the manner of §6, we construct the points and lines

$$p_1, r_1, p_2, r_2, \dots, p_{i-1}, r_{i-1},$$

and denote their totality by A , while the rest are collected under the symbol B , then the following is easily seen. The line p_i belongs to B ; for the point p_i lies on the lines r_{i-1} and p_i , of which the first belongs to A . Contrary to the hypothesis, it would already be completely determined if p_i also belonged to A . If now

$$p_i = t_k, \quad p_k = t_l,$$

and $k > i$, then the elements

$$t_k, p_{k+1}, r_{k+1}, \dots, p_n, r_n$$

with the exception of the point t_k all belong to A .

This is seen as follows. Suppose that a point t_j different from t_k , with $j > i$, belongs to B , and that $t_j = p_l$. Then $l > i$; $l = i$ is impossible, because $p_l = p_i$ would imply $t_j = t_k$. If $l > j$, then the line r_j , which contains the points t_j, p_j, p_{j+1} , would still not be completely determined after all preceding elements $p_1, r_1, \dots, p_j$ had been constructed. This is impossible, since p_i was supposed to be the last such element and r_j follows p_i . The only remaining possibility is $j > l > i$. Since r_j belongs to B and $r_j = p_l$, $j > l > i$, the sequence

$$t_i, p_{i+1}, r_{i+1}, p_{i+2}, \dots, p_n, r_n$$

contains before t_j an element different from t_k , namely p_l , that belongs to B .

Similarly, from the existence in the sequence $t_i, p_{i+1}, \dots, p_n, r_n$ of a line belonging to B one can infer the existence of a point, likewise belonging to B and different from t_k , which stands before the line in that sequence. Indeed, if $p_j = r_l$ is the line in question, then through p_j pass the lines r_{j-1}, p_j, r_j . Since $p_j = r_l$ is to occur in the sequence and belong to B , we first have $j > i$ and $l > i$. Since the point p_j is to be determined after the preceding elements have been constructed, because $j > i$, the line $p_j = r_l$ on which it lies must precede it in the normal order. Hence $j > l$. From $j > i$ it follows that p_j belongs to B ; from $j > l$ it follows that the point

$p_j = r_l$ stands before the line p_j in the sequence. Finally r_l cannot be identical with t_k , since $l = k$ would imply $j = i$, whereas $j > i$.

Thus every element that belongs to the sequence $t_i, p_{i+1}, \dots, p_n, r_n$, is different from t_k , and belongs to B , must be preceded in that sequence by an element with the same three properties. This gives a contradiction, since only finitely many elements are involved.

If the last condition of the configuration is now also taken into account, it follows from the foregoing that the elements

$$t_k, p_{k+1}, r_{k+1}, \dots, p_n, r_n, p_i, p_{i+1}, \dots$$

can be obtained in this order as joining lines and intersections of previously determined elements. For instance,

$$p_m r_n = r_n, (r_n, p_n) = p_n, \quad p_n r_{n-1} = r_{n-1}, \quad (r_{n-1}, p_{n-1}) = p_{n-1}, \dots$$

On the other hand, as soon as p_i has been assumed on the line r_{i-1} , one obtains $p_i r_i = r_i$, and analogously the following points and lines up to the element denoted by t_k .

The configuration still requires the points $t_k = p_i, p_k, p_{k+1}$, or respectively t_n, p_n, p_m if $k = n$, to lie on a line. Put first the joining line $p_{k+1} p_k = r'_k$ and the intersection $(r'_k, r_{i-1}) = p'_i$. As p_i traverses the line r_{i-1} , the point p'_i describes a projective point row on the same line. For all conditions of the configuration to be fulfilled, $p'_i = p_i$ must hold. Thus we arrive at the problem of finding the double elements of two incident point rows. It is well known that there are two such elements, and determining them usually requires a quadratic construction.

The cases in which p_i is identical with p_m , or in which the last element not completely determined by the preceding elements is a line, are dealt with in quite a similar way. A discussion of these cases may therefore be omitted.